

NAME:

ABBOTSLEIGH

4 UNIT MATHEMATICS

HSC ASSESSMENT # 2

March 1999

Time allowed: 1½ hours

Instructions to candidates

Attempt all questions.

Show all necessary working.

Approved calculators may be used.

Begin each question on a new page.

All questions are of equal value.

Question One (Start a new page)**Marks**

(a) Derive the equations of the tangent and normal at P(2,1) on the ellipse $4x^2 + 9y^2 = 25$.

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(b) Sketch the graph of $\frac{x^2}{16} - \frac{y^2}{9} = 1$, showing clearly the foci, directrices, asymptotes and vertices.

3

(c) In the expansion of $(1-2x)(1+ax)^{10}$ the coefficient of x^6 is 0. Find the value of a .

3

(d) (i) Evaluate $\int_0^1 \frac{dx}{(x+1)(x+3)}$

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(ii) Evaluate $\int_{-1}^2 x\sqrt{2-x} dx$ using $u^2 = 2-x$, for $u > 0$

Question Two (Start a new page)

(a) Assume $(3+2x)^{15} = \sum_{k=0}^{15} t_k x^k$

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(i) Use the Binomial Theorem to write an expression for t_k , $0 \leq k \leq 15$

(ii) Show that $\frac{t_{k+1}}{t_k} = \frac{2(15-k)}{3(k+1)}$

(iii) Hence, or otherwise, find the largest coefficient t_k (you may leave your answer in the form ${}^{15}C_k 3^a 2^b$)

(b) (i) Sketch the graphs of $y = x^2$ and $y = (x-2)^2$ on the same diagram.

3

(ii) Find the area enclosed between the curves $y = x^2$ and $y = (x-2)^2$ and the x-axis.

(c) Find the exact volume of the solid formed if the curve $y = \sqrt[4]{3x-1}$ is rotated about the x-axis from $x = 1$ to $x = 2$.

3

(d) Find $\int \sin^2 x \cos^2 x dx$

3

Question Three (Start a new page)

Marks

(a) $P(2at, at^2)$ is a point on the parabola $x^2 = 4ay$. S is the focus of the parabola. PQ is the perpendicular from P to the directrix d of the parabola. The tangent at P to the parabola cuts the axis of the parabola at the point R

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- (i) Show that the tangent at the point P to the parabola has equation $tx - y - at^2 = 0$
- (ii) Show that PR and QS bisect each other.
- (iii) Show that $PR \perp QS$
- (iv) State with reasons what type of quadrilateral PQRS is.

(b) (i) Find $\int \tan^4 x dx$

6

- (ii) Use the method of integration by parts to find a recurrence formula for $I_n = \int \sec^n x dx$

Question Four (Start a new page)

(a) (i) Derive the equation of the tangent at the point

6

$P(a \sec \theta, b \tan \theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

- (ii) The tangent at P meets the axes at Q and S. If OQRS is a rectangle, find the equation for the locus of R

(b) The points $P(2p, \frac{2}{p})$ and $Q(2q, \frac{2}{q})$ lie on the rectangular hyperbola $xy = 4$.

9

M is the midpoint of the chord PQ. P and Q move on the rectangular hyperbola so that the chord PQ always passes through the point R(4, 2).

- (i) Show that the chord PQ has equation $x + pqy = 2(p + q)$
- (ii) Show that $pq = p + q - 2$
- (iii) Hence show that the locus of M has equation

$$y = \frac{x}{x-2}$$

- (iv) On the same axes sketch the rectangular hyperbola $xy = 4$ and the locus of M, showing clearly the equations of any asymptotes and the point R.

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

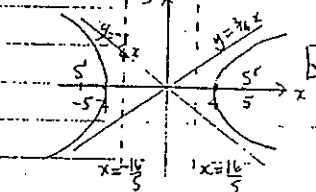
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question One (15 marks)

(a) $4x^2 + 9y^2 = 25$
 Differentiate w.r.t x
 $8x + 18y \frac{dy}{dx} = 0$
 $\frac{dy}{dx} = -\frac{8x}{18y}$
 At (2,1) $m = -\frac{8}{18}$
 Eqn of tangent is
 $y - 1 = -\frac{8}{9}(x - 2)$
 or $8x + 9y = 25$
 Eqn of normal is
 $y - 1 = \frac{9}{8}(x - 2)$
 or $9x - 8y = 10$

(b) $\frac{x^2}{16} - \frac{y^2}{9} = 1$
 $a = 4, b = 3, e = \frac{5}{4}$
 Foci are $(\pm 5, 0)$
 Directrices $x = \pm \frac{a^2}{c} = \pm \frac{16}{5}$
 Vertices $(\pm 4, 0)$
 Asymptotes $y = \pm \frac{3}{4}x$



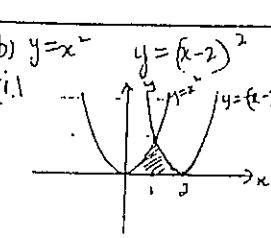
(c) $(1-2x)(1+ax)^{10}$
 $= (1-2x)(1 + 10ax + 45a^2x^2 + \dots + a^{10}x^{10})$
 Coeff $x^6 = 10a^5 - 20a^6 = 0$
 $10a^5 = 20a^6$
 $a = \frac{50}{210} = \frac{5}{21}$

(d) (i) $\int_0^1 \frac{dx}{(x+1)(x+3)}$
 Use method of partial fractions
 $\frac{1}{(x+1)(x+3)} = \frac{A}{x+1} + \frac{B}{x+3}$
 $1 = A(x+3) + B(x+1)$
 When $x = -1, A = \frac{1}{2}$
 When $x = -3, B = -\frac{1}{2}$
 $I = \frac{1}{2} \int_0^1 \frac{dx}{x+1} - \frac{1}{2} \int_0^1 \frac{dx}{x+3}$
 $= \frac{1}{2} [\ln(x+1) - \ln(x+3)]_0^1$
 $= \frac{1}{2} [\ln 2 - \ln 4 - \ln 1 + \ln 3]$
 $= \frac{1}{2} \ln \frac{3}{2}$

(ii) $\int_0^1 2\sqrt{2-x} dx, u = 2-x$
 When $x = 2, u = 0$ When $x = -1, u = 3$
 $I = \int_3^0 (2-u) u^{-1/2} (-du)$
 $= \int_0^3 (2-u) u^{-1/2} du$
 $= [4u^{1/2} - \frac{2}{3}u^{3/2}]_0^3$
 $= 4\sqrt{3} - \frac{2}{3} \cdot 3\sqrt{3} = \frac{8\sqrt{3}}{3}$

Question Two (15 marks)
 (a) $(3+2x)^{15} = \sum_{k=0}^{15} \binom{15}{k} 3^{15-k} 2^k x^k$
 $(i) (3+2x)^{15} = \sum_{k=0}^{15} \binom{15}{k} 3^{15-k} 2^k x^k$
 $(ii) \frac{t_{k+1}}{t_k} = \frac{\binom{15}{k+1} 3^{15-(k+1)} 2^{k+1}}{\binom{15}{k} 3^{15-k} 2^k}$
 $= \frac{15!}{k!(k+1)!} \times \frac{(k+1)!}{15!} \times \frac{2}{3} = \frac{2}{3} \times \frac{k+1}{15-k}$
 $= \frac{2(15-k)}{3(k+1)}$

(iii) $\frac{2}{3} \frac{(15-k)}{(k+1)} < 1$
 $3(15-k) < 3(k+1)$
 $27 - 3k < 3k + 3$
 $27 < 6k$
 $4.5 < k$
 $k = 6$
 $t_6 = 15C_6 3^9 2^6$



(ii) $A = \int_0^2 x^2 dx + \int_2^4 (x-2)^2 dx$
 $= [\frac{x^3}{3}]_0^2 + [\frac{(x-2)^3}{3}]_2^4$
 $= (\frac{8}{3} - 0) + (0 - \frac{-8}{3}) = \frac{16}{3}$
 $= \frac{16}{3} \text{ units}^2$

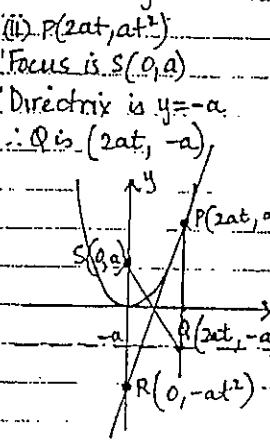
(c) $y = \sqrt{3x-1}$
 $V = \pi \int_0^9 y^2 dx$
 $= \pi \int_0^9 (3x-1) dx$
 $= \pi [\frac{3x^2}{2} - x]_0^9$
 $= \frac{2\pi}{9} (5\sqrt{5} - 2\sqrt{2}) u^3$

Question Two (cont)

(d) $\int \sin^2 x \cos^2 x dx$
 $= \int \frac{1}{4} \sin^2 2x dx$
 $= \int \frac{1}{8} (1 - \cos 4x) dx$
 $= \frac{1}{8} [x - \frac{\sin 4x}{4}] + C$
 $= \frac{x}{8} - \frac{\sin 4x}{32} + C$

Question Three (15 mks)

(i) $P(2at, at^2)$ $x^2 = 4ay$
 $y = \frac{x^2}{4a} \therefore \frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$
 At $P(2at, at^2)$
 $\frac{dy}{dx} = t$
 grad of tangent = t
 Eqn of tangent is
 $y - at^2 = t(x - 2at)$
 ie $tx - y - at^2 = 0$



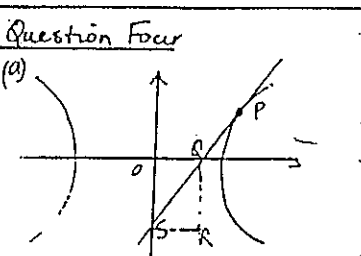
R is where $tx - y - at^2 = 0$
 intersects $x = 0$
 $\therefore R(0, -at^2)$

PR has midpt $(at, 0)$
 QS has midpt $(at, 0)$
 $\therefore PR + QS$ have same midpt
 $\therefore PR + QS$ bisect each other.

(iii) gradient PR = t.
 gradient QS = $-\frac{1}{t}$
 $\therefore t \times -\frac{1}{t} = -1 \therefore PR$ and QS are perpendicular to each other
 (iv) PQRS is a rhombus because the diagonals PR and QS bisect each other at 90° .

(b) (i) $\int \tan^4 x dx$
 $= \int \tan^2 x (\sec^2 x - 1) dx$
 $= \int (\tan^2 x \sec^2 x - \tan^2 x) dx$
 $= \int \tan^2 x \sec^2 x - \sec^2 x + 1 dx$
 $= \frac{\tan^3 x}{3} - \tan x + x + C$

$I_n = \int \sec^n x dx$
 $= \int \sec^{n-2} x \sec^2 x dx$
 So $u = \sec^{n-2} x$
 $\frac{du}{dx} = (n-2) \sec^{n-2} x \sec x \tan x$
 $\therefore I_n = \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x \sec x \tan x dx$
 $= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-1} x dx$
 $= \sec^{n-2} x \tan x - (n-2) I_{n-1}$
 $\therefore I_n = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}$
 $n > 1$



(i) The tangent at P is $\frac{x}{a} - \frac{y}{b} = 1$
 $\sec^2 x - \tan x = 1$
 It meets the x-axis at $A(\frac{1}{\sec^2 x}, 0)$. It meets the y-axis at $S(0, -\frac{1}{\tan x})$
 If PQRS is a rectangle the co-ordinates of R are $x = \frac{a}{\sec^2 x}, y = -\frac{b}{\tan x}$
 Hence $\sec^2 \theta = \frac{a}{x}, \tan \theta = -\frac{b}{y}$
 Since $\sec^2 \theta - \tan^2 \theta = 1$
 Then the locus of R is the curve $\frac{a^2}{x^2} - \frac{b^2}{y^2} = 1$

(b) The chord PQ has gradient $\frac{2-2}{-2-2} = \frac{0}{-4} = 0$
 and equation $y - 2 = 0$
 $y = 2$
 (ii) R(4,2) lies on PQ
 $\therefore (4) + p(2) = 2(p+q)$
 $4 + 2p = 2p + 2q$
 $4 = 2q$
 $q = 2$
 $\therefore M(\frac{1}{2}(2+2), \frac{1}{2}(\frac{2}{2} + \frac{2}{2})) = (2, 1)$

(iii) $M(\frac{1}{2}(2p+2q), \frac{1}{2}(\frac{2}{p} + \frac{2}{q}))$
 $x = p+q, y = \frac{p+q}{pq}$
 $\therefore y = \frac{p+q}{p+q-2} = \frac{x}{x-2}$

